



Overcurrent Fault Detector

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The overcurrent fault detector presented here is a general-purpose current-consumption sensor circuit that raises an alert when the flow of current through the system exceeds a certain limit. This simple circuit is based on ACS712 Hall Effect-based linear current sensor chip. The author's prototype is shown in Fig. 1.

Circuit and working

The circuit diagram of the overcurrent fault detector is shown in Fig. 1. It is built around ACS712 current sensor module, op-amp LM393 (IC1), diode 1N4007 (D1), pnp transistor BC557 (T1), 5V single-changeover relay (RL1) and a few other components.

A 5A version of the ACS712 current sensor is used here. The output sensitivity of the sensor designed for 5V single-supply operation is 185mV/A, and its quiescent output voltage (output of the sensor when the primary current is zero) nominally remains at $V_{cc}/2$.

The rest of the circuit is a common comparator built around one part of the popular LM393 (IC1). Analogue output of

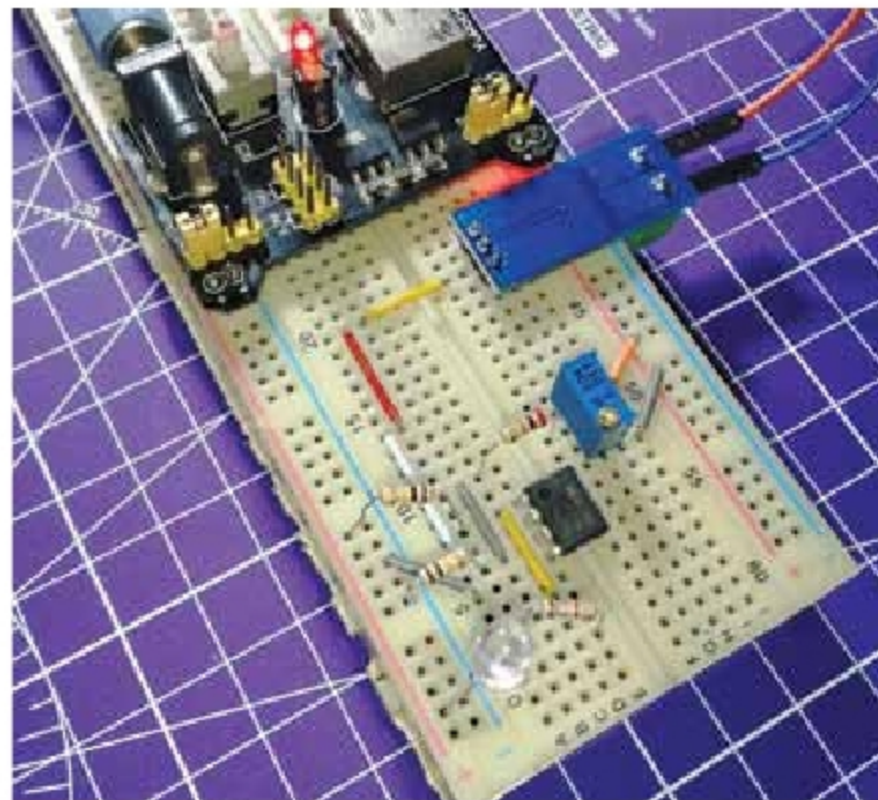


Fig. 1: Author's prototype wired on breadboard

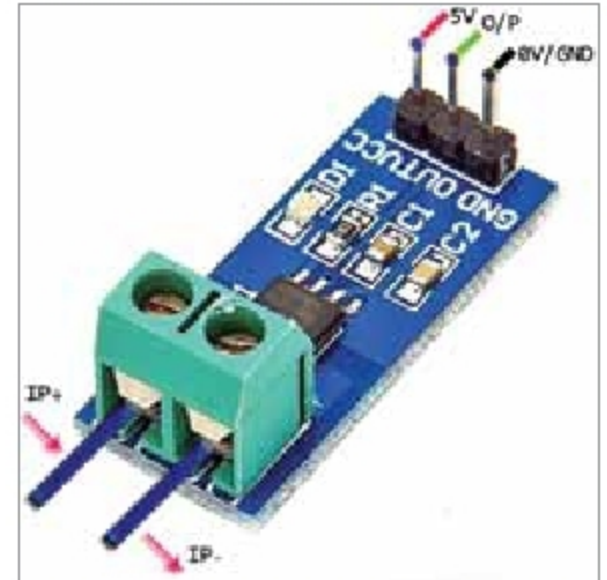


Fig. 3: ACS712 module

the ACS712 module is connected to inverting input (pin 2) of IC1, while non-inverting input (pin 3) of IC1 is connected to an adjustable potential divider (VR1, R2, and R3). A pull-up resistor (R4) and an overcurrent fault indicator (LED2) are connected at the output (pin 1) of IC1. This open-collector output pin 1 goes to a low (L) level when there is an overcurrent fault condition.

The ACS712 current sensor is available in surface mount SOIC8 package, which is often hard for some makers to handle and solder. Luckily, you can buy an expensive and compact pre-wired ACS712 module (Fig. 3) from most online component vendors. Since LM393 has an open-collector output, you can safely interface this circuit with a 5V or 3.3V microcontroller board through the CON3 connector. So, make arrangements in the circuit board to remove/add the optional output components (R4-R5-LED2) when required.

If you want to implement an add-on circuit, the easiest way is by using a simple electromagnetic relay driver circuit built around transistor T1, diode D1, and relay RL1. It can turn on appliances

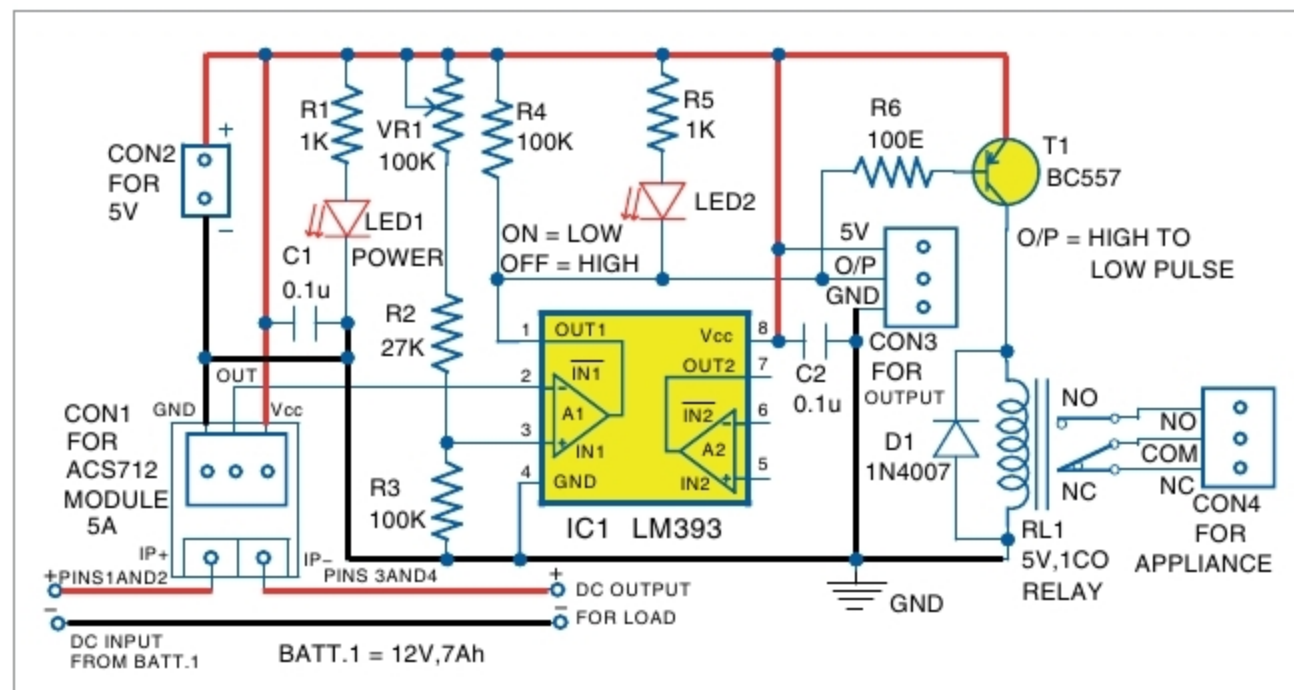


Fig. 2: Circuit diagram of the overcurrent fault detector